

李欣桐 (Xintong Li)

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教育背景

清华大学

2021年9月 – 至今

- 交叉信息研究院 (IIIS)
- 计算机科学博士研究生, 导师: 高鸣宇副教授

北京大学

2017年9月 – 2021年6月

- 计算机科学技术系
- 计算机科学学士, 导师: 孙广宇副教授

研究兴趣

我的主要研究兴趣是对包括稀疏性和动态性的不规则计算结构的软硬件协同优化。包括对稀疏计算算子(SpMSpM)的加速; 对近存计算硬件(Near-Data Processing / Processing-in-Memory)的加速; 和对大模型推理服务的加速 (特别是对于大模型推理服务中的稀疏性, 和复杂不均衡问题)。

主导项目 (第一作者)

[ISCA 2025] HYTE: 基于混合静态-动态方法的稀疏加速器灵活 Tiling

- 定义稀疏 Tiling 的完整设计空间, 分析不同稀疏特征的影响;
- 提出静态离线调度器, 通过轻量化采样寻找近似最优初始方案;
- 在运行时高效管理 Tiling 元数据, 并动态调整 Tile 形状以提升缓存利用率;
- 性能较现有方案提升 3.9–7.4 倍 (不同稀疏矩阵)。

[MICRO 2025] SeaCache: 适配稀疏加速器的高效自适应缓存

- 针对稀疏计算的可变访问特征, 提出两级地址映射, 减少 cacheline 浪费;
- 设计低成本预取策略, 利用一个矩阵的访问模式预测另一个矩阵访问;
- 探索数据、元数据与预取数据的最优缓存分配比例。

[进行中] AutoNDP: 面向大模型推理的近数据处理自动映射调度

- 支持NDP中的动态分配 Tile 至不同存储/计算位置, 提出针对 NMP 的 Tile 大小与形状选择方法;
- 构建高精度模拟器支持硬件设计空间探索。
- 对大模型推理服务中动态性的优化。(包括MoE负载均衡, session cache管理)

实习与工作经历

字节跳动 AML 异构计算组

2025年4月 – 至今

- 在新型硬件(存算芯片AiMx)上部署大模型推理服务, 评估新型硬件 (PIM、RRAM 等) 在大模型业务场景下的表现, 分析理论与实现差距;
- [校企合作] 完成 AutoNDP: 面向大模型推理的近数据处理自动映射调度的系统设计与调优。

论文发表

- SeaCache: Efficient and Adaptive Caching for Sparse Accelerators** MICRO, 2025 (CCF-A, TH-CPL-A)
Xintong Li, Jinchen Jiang, Mingyu Gao
- HYTE: Flexible Tiling for Sparse Accelerators via Hybrid Static-Dynamic Approaches** ISCA, 2025 (CCF-A, TH-CPL-A)
Xintong Li, Zhiyao Li, Mingyu Gao
- Adyna: Accelerating Dynamic Neural Networks with Adaptive Scheduling** HPCA, 2025 (CCF-A, TH-CPL-A)

Zhiyao Li, Bohan Yang, Jiaxiang Li, Taijie Chen, *Xintong Li*, Mingyu Gao

- **Hardware-Assistedservice Live Migration in Resource-Limited Edge Computing Systems**

Zhe Zhou, *Xintong Li*, Xiaoyang Wang, Zheng Liang, Guangyu Sun, Guojie Luo
- DAC, 2020 (CCF-A, TH-CPL-A)

荣誉与奖项

- 国家奖学金
 - 三好学生
 - 全国信息学奥林匹克竞赛(NOI) 银牌
- 2020
2019, 2020
2015

教学经历

- 课程助教

20470084 计算机体系结构，清华大学

维护并改进基于 RISC-V 的体系结构模拟器课程实验；组织习题课（课堂复习与内容拓展）。
- 2023秋季, 2024秋季

Xintong Li (李欣桐)

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Education

Tsinghua University

Sept. 2021 – Present

- Institute for Interdisciplinary Information Sciences (IIIS)
- Ph.D. student in Computer Science, advised by Prof. Mingyu Gao (高鸣宇).

Peking University

Sept. 2017 – June 2021

- School of Computer Science
- B.S. in Computer Science, advised by Prof. Guangyu Sun (孙广宇).

Research Interests

I specialize in computational acceleration, with a particular focus on accelerating sparse computations. This involves analyzing the impact of different sparse features on computational aspects such as tiling; scheduling; buffer management, and optimizing for specific sparse characteristics. Additionally, I am interested in exploring other computational models, such as near-data processing (NDP).

Leading Projects

[ISCA 2025] HYTE: Flexible Tiling for Sparse Accelerators via Hybrid Static-Dynamic Approaches

- Research Problem: Sparse features significantly impact the effectiveness and overhead of tiling in sparse tensor algebra. The optimal tile size; tile shape; inter-tile order; and buffer allocation vary across different sparse matrices. Additional hardware support is also required to explore the full design space.
- Define the full design space of sparse tiling and explore the influences of different sparse features.
- Propose a static offline scheduler to identify a near-optimal initial tiling scheme through effective and lightweight sampling.
- Supports runtime efficient management of the tiling metadata in both the off-chip memory and the on-chip buffer, as well as a technique of dynamic tuning on the tile shape to ensure high buffer utilization in the presence of highly varying data patterns.
- Outperforms state-of-the-art sparse tiling strategies by $3.9\times$ to $7.4\times$ on average for diverse sparse matrices.

[MICRO 2025] SeaCache: Efficient and Adaptive Caching for Sparse Accelerators

- Research Problem: Sparse accelerators introduce new features, including variable fiber length and distribution; the need for random indexing and eviction; and irregular yet predictable access patterns. Conventional cache designs in general-purpose hardware or dense accelerators are unable to accommodate these new features.
- Design a two-part address mapping method to avoid cacheline underutilization or high indirect access costs.
- Consider the trade-off between prefetching costs and hit rate improvements, and design a prefetch data management strategy with lower costs. (Sparse computations can leverage prefetching from one matrix to predict the future access patterns of another matrix, thereby achieving better replacement and improving cache hit rates.)
- Explore the optimal buffer allocation ratio among the two-part data, metadata, and prefetch data.

[In Progress] AutoNDP: Automating Computation Scheduling in Near-Data Processing Systems for LLM Inference

- Develop a hybrid mapping strategy combining compute-centric mapping (mapping operators to computational units, which leads to DRAM bank capacity imbalance and additional communication in NMP scenarios) and data-centric mapping (mapping data such as weights and KV cache to DRAM banks, which causes computational unit imbalance). The specific implementation dynamically maps each tile (divided operators) based on runtime conditions.
- During simulation, store the performance of operators with specific shapes on specific hardware, simulating operators of the same shape only once. This avoids the inaccuracy of analytical models and the slow speed of

cycle-accurate models.

- In implementation, incorporate tile size and shape selection for NMP scenarios, allowing the same operator to be stored/computed in different locations; dynamically allocated tiles based on runtime conditions; enhanced precise simulation and evaluation of softmax; used a simulator integrating the above three components for hardware Design Space Exploration (DSE).

Experience

Heterogeneous Computing Group, ByteDance

April 2025 - Present

- Evaluate and analyze the performance of new hardware technologies, such as PIM and RRAM, on business models, analyzing the gap between theoretical and actual implementations.
- Complete the autoNDP project.

Publications

- **SeaCache: Efficient and Adaptive Caching for Sparse Accelerators** MICRO, 2025
Xintong Li, Jincheng Jiang, Mingyu Gao
- **HYTE: Flexible Tiling for Sparse Accelerators via Hybrid Static-Dynamic Approaches** ISCA, 2025
Xintong Li, Zhiyao Li, Mingyu Gao
- **Adyna: Accelerating Dynamic Neural Networks with Adaptive Scheduling** HPCA, 2025
Zhiyao Li, Bohan Yang, Jiayang Li, Taijie Chen, *Xintong Li*, Mingyu Gao
- **Hardware-Assisted Service Live Migration in Resource-Limited Edge Computing Systems** DAC, 2020
Zhe Zhou, *Xintong Li*, Xiaoyang Wang, Zheng Liang, Guangyu Sun, Guojie Luo

Honors and Awards

- National Scholarship 2020
- Honored Student 2019, 2020
- National Olympiad in Informatics (NOI), Silver Medal 2015

Teaching

- Teaching Assistant Fall 2023, Fall 2024
20470084 Computer Architecture, Tsinghua University
Maintained and improved a course project focused on a computer architecture simulator for RISC-V.
Taught review session classes on review and expansion.